MiniOS Alpha -- User Manual

Operating System: MiniOS Alpha | Project: IT 3131 Final Project

1. Introduction

MiniOS Alpha is a simple command-line operating system simulator written in Python. It runs inside any terminal and simulates fundamental OS components (Command shell, hierarchical file management, process lifecycle tracking, memory allocation, role-based user management, error handling.) It is a simulator designed to demonstrate operating systems concepts, not a real bootable operating system.

2. System Requirements

- Operating System: Windows (Command Prompt / CMD recommended), macOS, or Linux.
- Python Version: Python 3.8 or higher.
- Dependencies: None (uses standard Python library only: os, sys, time, datetime, json, hashlib, getpass, platform).

3. Installation

- 1. Download or clone the project repository:

```
git clone https://github.com/matthewsaint/minios-alpha.git
cd minios-alpha
```

- 2. Verify Python is installed and accessible:

```
python --version
```

- 3. Confirm core project files exist in the directory (main.py, filesystem.py, processes.py, memory.py, users.py, requirements.txt).

4. How to Start MiniOS

- 1. Open Windows Command Prompt (CMD) and navigate to the project directory.
- 2. Launch the simulator:

```
python main.py
```

- 3. MiniOS displays the boot banner and prompts for login:

```
========================================
          MINI OS ALPHA v1.0
========================================

Welcome to MiniOS!

========================================
              MiniOS Login
========================================
Demo accounts: admin / admin123, student / student123

Username:
```

- 4. After logging in, the shell prompt appears:

```
MiniOS>
```

5. Login Instructions

MiniOS Alpha requires authentication before granting shell access.

Default Accounts

Username	Password	Role
admin	admin123	Administrator
student	student123	Standard User

Login Steps

- 1. Enter admin or student at the Username: prompt and press Enter.

- 2. Enter the password at the Password: prompt.

(Note: Password characters are hidden in CMD for security.)

- 3. On successful login, the welcome message is displayed:

```
   Login successful.

   Welcome, admin!
   Type 'help' to see the available commands.

   MiniOS>
```

- 4. Attempt Limit: The system allows up to 3 consecutive attempts. After 3 failed attempts, MiniOS shuts down automatically.
- 5. Switching Users: Use logout or login to return to the login screen.

6. Command List

MiniOS Alpha provides 24 commands and aliases. Each command is documented below using the required reference format.

6.1 General Commands

COMMAND: help

Purpose: Displays the list of all available commands organized by category.
Syntax: **`help`**

Example:

```
MiniOS> help
Available Commands:

[General]
  help                  Show this list
  clear                 Clear the screen
  date / time           Show the date / the time
  whoami                Show the current user
  systeminfo            Show system information
  history               Show commands typed this session
  echo <text>           Print text
  version               Show the MiniOS version
  exit                  Shut down MiniOS

[Files]
  dir [path]            List files and folders
  mkdir <name>          Create a directory
  create <file>         Create an empty file
  write <file> <text>   Write text into a file
  read <file>           Show a file's contents
  delete <name>         Delete a file or empty directory
  rename <old> <new>    Rename a file or directory
  cd <path>             Change directory (cd .. goes up)
  pwd                   Show the current directory

[Processes]
  run <name>            Start a simulated process
  process               List running processes
  kill <pid>            Terminate a process

[Memory]
  memory                Show memory status
  allocate <MB>         Allocate memory
  free <MB>             Release memory

[Users]
  users                 List user accounts
  logout                Log out and return to login
  login                 Switch user (alias of logout)
```

COMMAND: clear

Purpose: Clears the terminal screen. (`cls` is supported as an alias).
Syntax: **`clear`**

Example:

```
MiniOS> clear
```

COMMAND: date

Purpose: Displays the current system date.
Syntax: **`date`**

Example:

```
MiniOS> date
Current Date: Thursday, October 22, 2026
```

COMMAND: time

Purpose: Displays the current system time in 12-hour AM/PM format.
Syntax: **`time`**

Example:

```
MiniOS> time
Current Time: 09:45:10 PM
```

COMMAND: whoami

Purpose: Displays the username of the current logged-in user.
Syntax: **`whoami`**

Example:

```
MiniOS> whoami
Current User: admin
```

COMMAND: systeminfo

Purpose: Displays simulated hardware, operating system, and session information.
Syntax: **`systeminfo`**

Example:

```
MiniOS> systeminfo
========================================
           SYSTEM INFORMATION
========================================

OS Name:      MiniOS Alpha
Version:      1.0
Architecture: 64-bit
CPU:          Virtual CPU
Memory:       1024 MB (256 MB used)
Host:         Windows (Python 3.11.4)
User:         admin (Administrator)
Processes:    0 running
Files/Dirs:   0 file(s), 2 folder(s)
Uptime:       1m 45s

========================================
```

COMMAND: history

Purpose: Shows the history of commands entered during the current session.
Syntax: **`history`**

Example:

```
MiniOS> history
  1  whoami
  2  dir
  3  systeminfo
  4  history
```

COMMAND: echo

Purpose: Prints the specified text back to the screen.
Syntax: **`echo <text>`**

Example:

```
MiniOS> echo Hello MiniOS
Hello MiniOS
```

COMMAND: version

Purpose: Displays the MiniOS version.
Syntax: **`version`**

Example:

```
MiniOS> version
MiniOS Alpha v1.0
```

COMMAND: exit

Purpose: Shuts down the MiniOS simulator. (`shutdown` is supported as an alias).
Syntax: **`exit`**

Example:

```
MiniOS> exit
Shutting down MiniOS...

Goodbye!
```

6.2 File Management Commands

COMMAND: pwd

Purpose: Shows the current working directory path.
Syntax: **`pwd`**

Example:

```
MiniOS> pwd
\
```

COMMAND: dir

Purpose: Lists files and directories in the current folder or specified path. (`ls` is supported as an alias).
Syntax: **`dir [path]`**

Example:

```
MiniOS> dir
Directory of \

     <DIR>  Documents
        12  notes.txt
```

COMMAND: mkdir

Purpose: Creates a new directory.
Syntax: **`mkdir <folder>`**

Example:

```
MiniOS> mkdir Documents
Directory created.
```

COMMAND: create

Purpose: Creates a new empty file.
Syntax: **`create <file>`**

Example:

```
MiniOS> create notes.txt
File created.
```

COMMAND: write

Purpose: Writes text into a file. Creates the file if it does not exist.
Syntax: **`write <file> <text>`**

Example:

```
MiniOS> write notes.txt Operating Systems Final Project
File saved.
```

COMMAND: read

Purpose: Displays the contents of a file.
Syntax: **`read <file>`**

Example:

```
MiniOS> read notes.txt
Operating Systems Final Project
```

COMMAND: delete

Purpose: Deletes a file or an empty directory.
Syntax: **`delete <name>`**

Example:

```
MiniOS> delete notes.txt
File deleted.
```

COMMAND: rename

Purpose: Renames an existing file or directory.
Syntax: **`rename <old> <new>`**

Example:

```
MiniOS> rename notes.txt draft.txt
Renamed.
```

COMMAND: cd

Purpose: Changes the current directory. (`cd ..` goes to parent directory; `cd \` returns to root).
Syntax: **`cd <path>`**

Example:

```
MiniOS> cd Documents
Current directory: \Documents

MiniOS> cd ..
Current directory: \
```

6.3 Process Management Commands

COMMAND: run

Purpose: Starts a new simulated process and assigns it a PID.
Syntax: **`run <name>`**

Example:

```
MiniOS> run calculator
Process started.

PID: 1
Name: calculator
Status: RUNNING
```

COMMAND: process

Purpose: Lists all currently running simulated processes.
Syntax: **`process`**

Example:

```
MiniOS> process
PID    NAME            STATUS    STARTED
1      calculator      RUNNING   21:45:00
```

COMMAND: kill

Purpose: Terminates a simulated process using its PID.
Syntax: **`kill <pid>`**

Example:

```
MiniOS> kill 1
Process 1 terminated.
```

6.4 Memory Management Commands

COMMAND: memory

Purpose: Displays current memory status including total, used, and free memory.
Syntax: **`memory`**

Example:

```
MiniOS> memory
Memory Status
------------------------
Total Memory: 1024 MB
Used Memory:   256 MB
Free Memory:   768 MB
```

COMMAND: allocate

Purpose: Allocates a specified amount of memory in megabytes (MB).
Syntax: **`allocate <MB>`**

Example:

```
MiniOS> allocate 200
200 MB allocated.
```

COMMAND: free

Purpose: Releases a specified amount of user-allocated memory in megabytes (MB).
Syntax: **`free <MB>`**

Example:

```
MiniOS> free 200
200 MB released.
```

6.5 User Management Commands

COMMAND: users

Purpose: Lists all registered user accounts, their roles, and identifies the active user.
Syntax: **`users`**

Example:

```
MiniOS> users
USER        ROLE
admin       Administrator  <- you
student     Standard User
```

COMMAND: logout

Purpose: Logs out the current user and returns to the login screen. (`login` is an alias).
Syntax: **`logout`**

Example:

```
MiniOS> logout
Logging out admin...

========================================
              MiniOS Login
========================================
Demo accounts: admin / admin123, student / student123

Username:
```

7. Common Errors

MiniOS Alpha handles invalid inputs gracefully and does not crash when an error occurs. Below are the most common error messages, their causes, and resolutions:

- 1. Unknown Command
- _Input:_ hellooo
- _Message:_

```
    Unknown command.

    Type 'help' to see the available commands.
```

- _Resolution:_ Verify spelling or type help to view valid commands.
- 2. File Does Not Exist
- _Input:_ read missing.txt
- _Message:_ Error: File does not exist.
- _Resolution:_ Use dir to verify the file exists in the current directory.

- 3. Entity Already Exists
- _Input:_ mkdir Documents (when Documents already exists)
- _Message:_ Error: A file or directory with that name already exists.
- _Resolution:_ Choose a different name or delete the existing entity.
- 4. Directory Not Empty
- _Input:_ delete Documents (folder has files inside)
- _Message:_ Error: Directory is not empty.
- _Resolution:_ Delete the files inside the directory before deleting the directory.
- 5. Invalid File/Directory Name
- _Input:_ create report:final
- _Message:_ Error: Invalid name. Avoid / \ : * ? " < > |
- _Resolution:_ Remove forbidden special characters from the name.
- 6. Process Does Not Exist
- _Input:_ kill 999
- _Message:_ Error: Process does not exist.
- _Resolution:_ Use process to find valid running PIDs.
- 7. Process Limit Reached
- _Input:_ run app (when 10 processes are already running)
- _Message:_ Error: Process limit reached (10). Kill one first.
- _Resolution:_ Use kill <pid> to terminate an existing process before starting a new one.
- 8. Insufficient Free Memory
- _Input:_ allocate 1000 (only 768 MB free)
- _Message:_ Error: Not enough free memory. Only 768 MB available.
- _Resolution:_ Check memory and request an amount less than or equal to free memory.
- 9. Cannot Free Unallocated Memory
- _Input:_ free 50 (when 0 MB has been allocated by the user)
- _Message:_ Error: Cannot free 50 MB. Only 0 MB is allocated by you.
- _Resolution:_ You can only free memory that you previously allocated.
- 10. Missing Command Arguments
- _Input:_ write notes.txt
- _Message:_ Error: Missing argument. Usage: write <file> <text>
- _Resolution:_ Supply all required parameters indicated by the usage hint.

8. How to Exit MiniOS

Method 1: The `exit` Command

Type exit or shutdown at the prompt:

```
MiniOS> exit
Shutting down MiniOS...

Goodbye!
```

Method 2: Keyboard Interrupts

Press Ctrl+C or Ctrl+D (or Ctrl+Z on Windows). The system intercepts the interrupt and exits gracefully:

```
Shutting down MiniOS...

Goodbye!
```

State Persistence

- Filesystem: Files and folders persist across reboots via minios_disk.json.
- Processes and Memory: Simulated processes and dynamic memory allocations are volatile and reset cleanly to default upon restart.